

Lab Report
On
“CLOUD COMPUTING”

In partial fulfillment of the requirements for the degree of
Bachelor of Computer Applications

Submitted by:

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Under the esteemed guidance of

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Table of Contents

Acknowledgement	2
Lab 1: Getting Started with Azure	4
Lab 2: Azure Storage Services	7
Lab 3: Azure Virtual Machines	10
Lab-4 NGIX Configuration.....	14
Lab-5 Virtualization	23
Lab-6 Installing Linux Mint on VMware Workstation	29

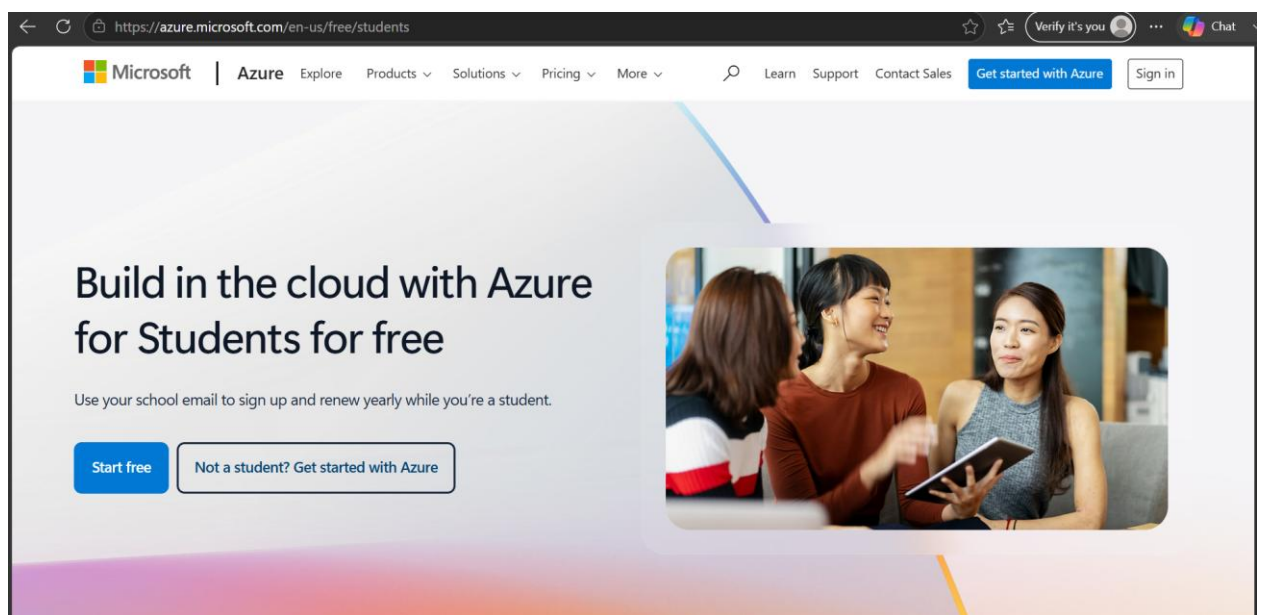
Lab 1: Getting Started with Azure

Objective: To understand the Azure Portal, subscription model, resource groups, and basic cloud services.

Tasks:

1. Create an Azure Account

- Visit: <https://azure.microsoft.com/en-us/free/students>



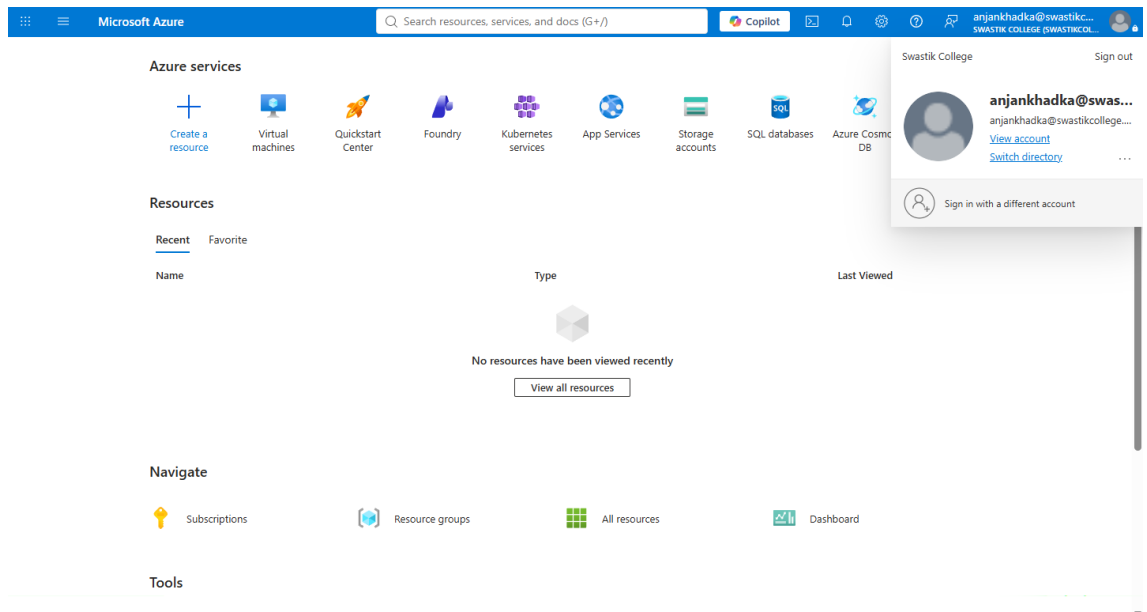
Click Start Free and sign in with a Microsoft account (use a school/university email).

- Select Student Option to get \$100 credits via [Azure for Students](#).
- Complete the sign up and verification (phone number + identity verification).

Note: No credit card is needed for student accounts.

2. Explore Azure Portal and Cloud Shell

- Login to the [Azure Portal](#).



- Familiarize yourself with:
 - Dashboard
 - Resource Groups
 - All Services
 - Marketplace
- Click on Cloud Shell (top-right menu bar → terminal icon)
 - Choose Bash (or PowerShell)
 - Set up Cloud Shell storage if prompted

Tip: Cloud Shell comes preinstalled with Azure CLI.

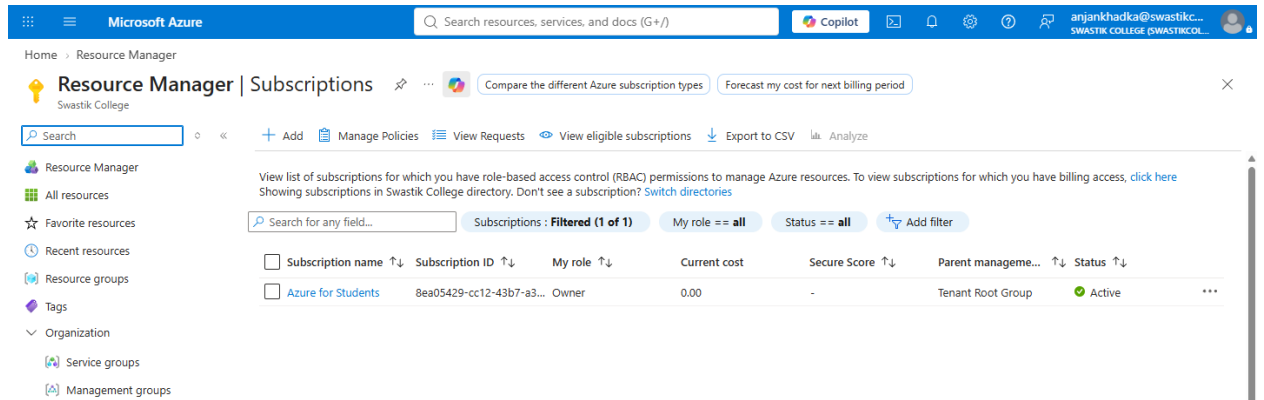
3. Create a Resource Group

A resource group is a logical container for your Azure resources.

- **Via Portal:**
 - Go to Resource Groups → Create
 - Resource Group Name: **lab1-rg**
 - Region: **East US** (or nearest region)

4. Verify Azure Subscription

- Go to Subscriptions and verified the Azure for students subscription.
- Confirmed that the subscription status was Active.



The screenshot displays the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and user information for 'anjankhadka@swastik...'. The main header shows 'Resource Manager | Subscriptions' for 'Swastik College'. A left-hand navigation pane lists various resource management options. The main content area shows a list of subscriptions, with one subscription, 'Azure for Students', listed as 'Active'.

Subscription name	Subscription ID	My role	Current cost	Secure Score	Parent manageme...	Status
Azure for Students	8ea05429-cc12-43b7-83...	Owner	0.00	-	Tenant Root Group	Active

Lab 2: Azure Storage Services

Objective:

To Learn to create and manage Azure Blob Storage, including uploading files and accessing them via public URLs using both the Portal and Azure CLI.

Via Portal:

- Go to Storage accounts → Create

Microsoft Azure

Search resources, services, and docs (G+/)

Home > Storage center | Blob Storage

Create a storage account

Azure Storage is a Microsoft-managed service providing cloud storage that is highly available, secure, durable, scalable, and redundant. Azure Storage includes Azure Blobs (objects), Azure Data Lake Storage Gen2, Azure Files, Azure Queues, and Azure Tables. The cost of your storage account depends on the usage and the options you choose below.
[Learn more about Azure storage accounts](#)

Project details

Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription *

Azure for Students

Resource group *

lab-1

[Create new](#)

Instance details

Storage account name * ⓘ

lab1anjan

Region * ⓘ

(Asia Pacific) Central India

[Deploy to an Azure Extended Zone](#)

Primary service *

Azure Blob Storage or Azure Data Lake Storage

This helps us provide relevant guidance. It doesn't restrict your storage to this resource type. [Learn more](#)

Performance * ⓘ

Standard: Recommended for most scenarios (general-purpose v2 account)

Premium: Recommended for scenarios that require low latency.

Redundancy * ⓘ

Geo-redundant storage (GRS)

Make read access to data available in the event of regional unavailability.

Geo priority replication guarantees Blob storage data is geo-replicated within 15 minutes.

Previous

Next

Review + create

a) Fill in the details:

- Subscription: (Select your active subscription)

- Resource Group: (or create a new one)
- Storage Account Name: (must be globally unique) [lab1anjan]
- Region: East US (or your preferred region) [(Asia Pacific) Central India]
- Performance: Standard
- Redundancy: Locally-redundant storage (LRS)

b) Click Review + Create, then Create

The screenshot shows the Microsoft Azure portal interface. At the top, there's a navigation bar with 'Microsoft Azure' and a search bar. Below it, the 'Storage center | Blob Storage' header is visible. The left sidebar contains a navigation menu with options like 'Overview', 'All storage resources', 'Object storage', 'File storage', 'Block storage', 'Data management', 'Migration', 'Partner solutions', 'Management services', and 'Help'. The main content area is titled 'Resources' and displays a table of storage accounts. The table has columns for Name, Type, Kind, Resource Group, Location, Subscription, and Defender for storage. One storage account is listed: 'lab1anjan' of type 'Storage account' and kind 'StorageV2', located in 'Central India' under the 'Azure for Students' subscription.

Name	Type	Kind	Resource Group	Location	Subscription	Defender for sto...
lab1anjan	Storage account	StorageV2	lab-1	Central India	Azure for Students	Not protected

2. Create a Blob Container

Via Portal:

1. Navigate to your newly created Storage Account
2. Under Data storage, select Containers
3. Click + Container

Search

- Overview
- Activity log
- Tags
- Diagnose and solve problems
- Access Control (IAM)
- Data migration
- Events
- Storage browser
- Storage Mover
- Partner solutions
- Resource visualizer
- ▼ Data storage
 - Containers**
 - Classic file shares
 - Queues
 - Tables
- > Security + networking
- > Data management

+ Add container ↑ Upload Refresh Delete ...

Search containers by prefix

Only show active containers

Showing all 1 items

☐	Name	Last modified	Anonymous access level
☐	 \$logs	24/09/2026, 20:47:10	Private

Lab 3: Azure Virtual Machines

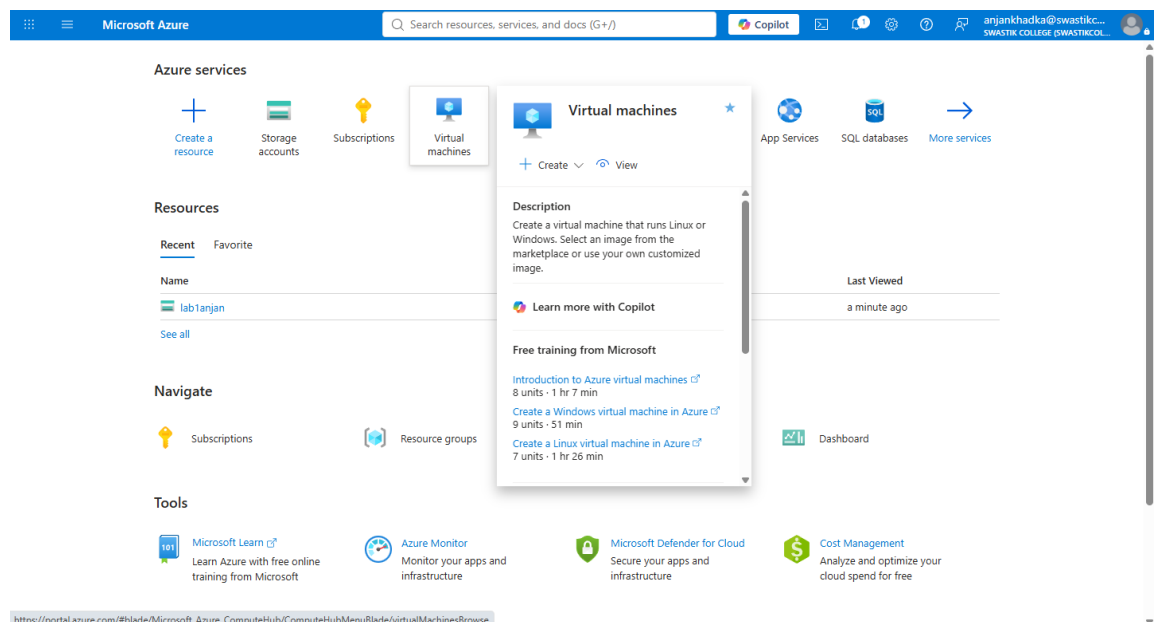
Objective:

To Deploy a Linux virtual machine on Azure, connect to it via SSH, and host a static website using Apache or Nginx.

1. Create a Linux-based VM

Via Portal:

- Go to Virtual Machines → Create



1. Fill in the following:

- Subscription: Your active one
- Resource Group: **lab-1**
- VM name: **anjan-vm**
- Region: East US (or nearby)

All services

Create a virtual machine ...

Basics

Disks

Networking

Management

Monitoring

Advanced

Tags

Review + create

Subscription * ⓘ

Azure for Students



Resource group * ⓘ

lab-1

[Create new](#)

Instance details

Virtual machine name * ⓘ

anjan-vm

Region * ⓘ

(Asia Pacific) East Asia

[Deploy to an Azure Extended Zone](#)

Availability options ⓘ

Availability zone

Zone options * ⓘ



Self-selected zone

Select a specific zone for your VM



Azure-selected zone (Preview)

Let Azure assign the best zone for your needs

Availability zone * ⓘ

Zone 2

Security type ⓘ

Trusted launch virtual machines

All services

Create a virtual machine ...

Basics

Disks

Networking

Management

Monitoring

Advanced

Tags

Review + create

Subscription * ⓘ

Security type ⓘ

Trusted launch virtual machines

[Configure security features](#)

Image * ⓘ

 Ubuntu Server 24.04 LTS - x64 Gen2

[See all images](#) | [Configure VM generation](#)

VM architecture * ⓘ



Arm64



x64

Size * ⓘ

Standard_B2as_v2 - 2 vcpus, 8 GiB memory (US\$76.65/month)

[See all sizes](#) | [Customize cores](#)

Run with Azure Spot discount ⓘ



Administrator account

Authentication type * ⓘ



SSH public key

Authentication type

☒ Password

Username * ⓘ

Password * ⓘ

Confirm password * ⓘ

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ⓘ ☐ None ☒ Allow selected ports

Select inbound ports *

ⓘ This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

Previous
Next
Review + create

- After deployment, go to the VM → Note down the Public IP address

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and user information. The main content area is titled 'Compute infrastructure | Virtual machines'. On the left, there is a sidebar with navigation options like Overview, All resources, Infrastructure, and Virtual machines. The main area displays a table of virtual machines. The table has columns for Name, Subscription, Resource Group, Location, Status, Operating system, Size, Public IP address, and Disks. One VM is listed: 'anjan-vm' under the 'Azure for Students' subscription, in the 'lab-1' resource group, located in 'East Asia', with a status of 'Running', operating on 'Linux', and having a size of 'Standard_B2as_v1'. Its public IP address is '40.81.17.223' and it has 1 disk.

Name	Subscription	Resource Group	Location	Status	Operating system	Size	Public IP address	Disks
anjan-vm	Azure for Students	lab-1	East Asia	Running	Linux	Standard_B2as_v1	40.81.17.223	1

```
Anjan@anjan-vm: ~
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Anjan> ssh Anjan@40.81.17.223
The authenticity of host '40.81.17.223 (40.81.17.223)' can't be established.
ED25519 key fingerprint is SHA256:w8rLLowFARsf54+n5SmNG+HngubGrhzthnyJlX3N3Uc.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '40.81.17.223' (ED25519) to the list of known hosts.
Anjan@40.81.17.223's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1022-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Sep 24 15:28:06 UTC 2026

System load:  0.08               Processes:            122
Usage of /:   5.6% of 28.02GB    Users logged in:     0
Memory usage: 3%                IPv4 address for eth0: 172.16.0.4
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

Anjan@anjan-vm:~$ |
```

Lab-4 NGIX Configuration

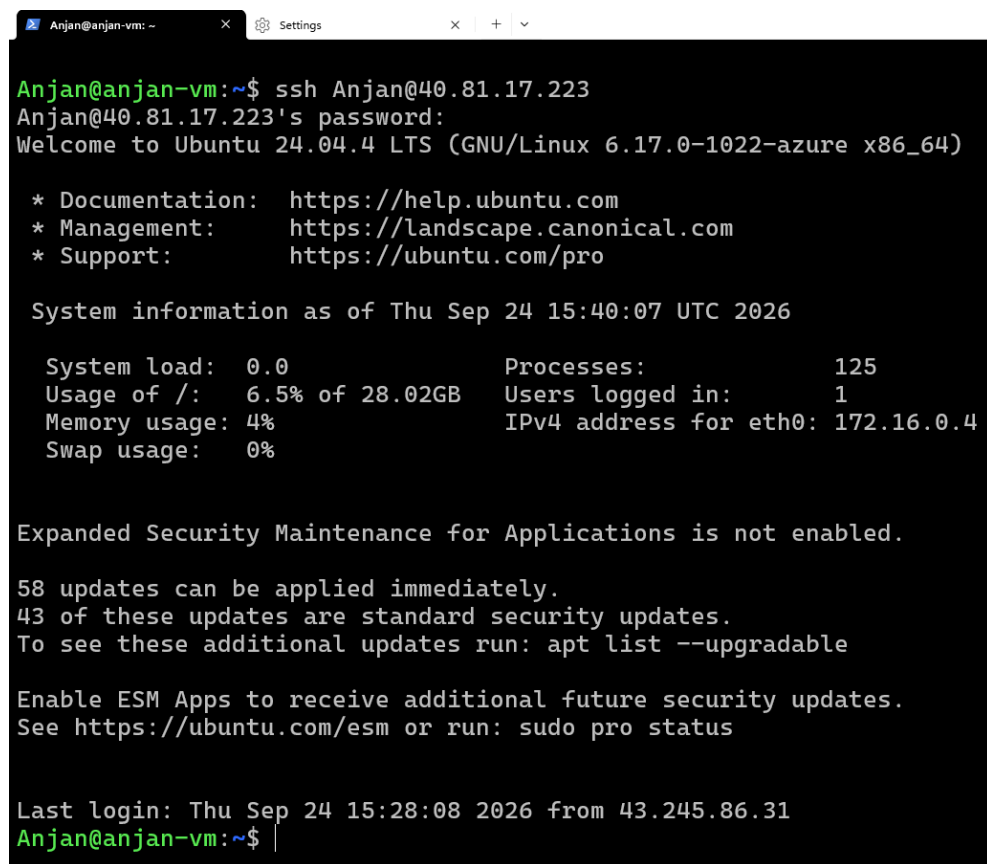
1. Connect to the Virtual Machine

- Connected to the Azure VM using SSH.
- Verified successful access to the Linux environment.

ssh <host-name>@<ip-address>

ssh Anjan@40.81.17.223

bcA-laB12345



```
Anjan@anjan-vm: ~$ ssh Anjan@40.81.17.223
Anjan@40.81.17.223's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1022-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Sep 24 15:40:07 UTC 2026

System load:  0.0               Processes:            125
Usage of /:   6.5% of 28.02GB    Users logged in:     1
Memory usage: 4%               IPv4 address for eth0: 172.16.0.4
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

58 updates can be applied immediately.
43 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Sep 24 15:28:08 2026 from 43.245.86.31
Anjan@anjan-vm:~$ |
```

2. Install NGINX

- Updated the system packages.
- Installed NGINX and verified that the service was running successfully.

sudo apt update

```
Anjan@anjan-vm:~$ sudo apt update
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
52 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

sudo apt install nginx -y

```
Anjan@anjan-vm:~$ sudo apt install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nginx is already the newest version (1.24.0-2ubuntu7.18).
0 upgraded, 0 newly installed, 0 to remove and 52 not upgraded.
```

sudo systemctl status nginx

```
Anjan@anjan-vm:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-09-24 15:32:08 UTC; 10min ago
     Docs: man:nginx(8)
   Process: 2170 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=0)
   Process: 2172 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0)
  Main PID: 2201 (nginx)
    Tasks: 3 (limit: 9432)
   Memory: 2.4M (peak: 5.2M)
      CPU: 21ms
   CGroup: /system.slice/nginx.service
           └─2201 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─2204 "nginx: worker process"
               └─2205 "nginx: worker process"

Sep 24 15:32:08 anjan-vm systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server:
Sep 24 15:32:08 anjan-vm systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server:
lines 1-17/17 (END)
```

3. Connect GitHub

- Generated an SSH key on the virtual machine.

ssh-keygen -t ed25519 -C

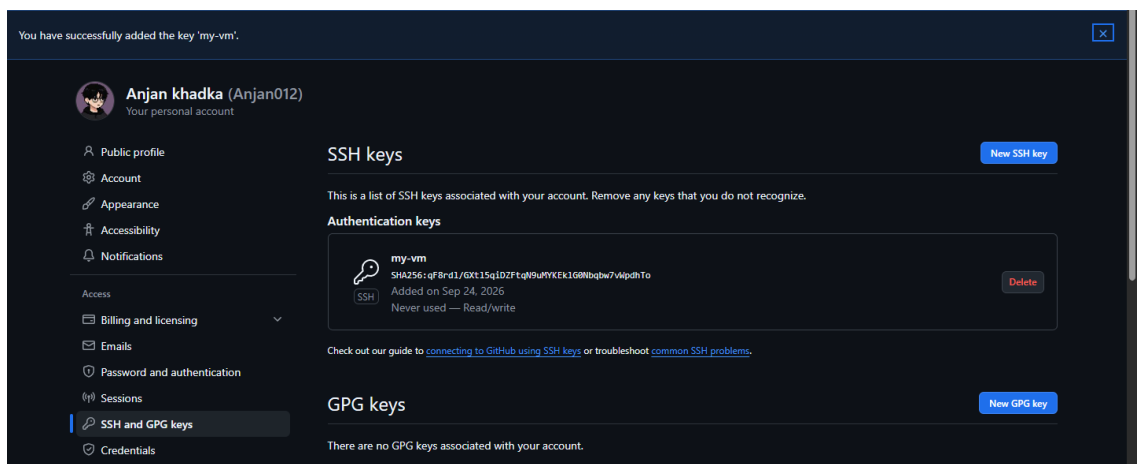
```
Anjan@anjan-vm:~$ ssh-keygen -t ed25519 -C "anjankhadka199@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/Anjan/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/Anjan/.ssh/id_ed25519
Your public key has been saved in /home/Anjan/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:qF8rdL/GXtL5qiDZftqN9uMYKEkLG0Nbqbw7vWpdhTo anjankhadka199@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|
| . . .
| . o .
| .=.. .
| oB . .
| +.S...
|
| o..E*. = o.
| . o+*o0 = o.o
| .*o+= 0.. ..
|
| oo=ooo.=o.
|
+-----[SHA256]-----+
```

- View and copy your public key:

```
cat ~/.ssh/id_ed25519.pub
```

```
Anjan@anjan-vm:~$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIBM3P77bkKEq7teAMNF6Zo7LMh/zMX/2VDD5DqJSbSLS anjankhadka199@gmail.com
Anjan@anjan-vm:~$ |
```

- Add it to GitHub → Settings → SSH and GPG Keys → New SSH Key.



- Tested the SSH connection with GitHub:

```
ssh -T git@github.com
```

```
Anjan@anjan-vm:~$ ssh -T git@github.com
The authenticity of host 'github.com (20.205.243.166)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Hi Anjan012! You've successfully authenticated, but GitHub does not provide shell access.
Anjan@anjan-vm:~$ |
```

4. Clone the Project

- Navigate to the directory where you want the project:

```
cd /var/www/
```

```
Anjan@anjan-vm:~$ cd /var/www/
```

- Cloned the required project repository from GitHub.

```
sudo git clone git@github.com:username/repository.git
```

- Configured the required permissions for the project directory.

```
sudo chown -R bca-lab:bca-lab /var/www
```

```
Anjan@anjan-vm:/var/www$ sudo git clone git@github.com:Anjan012/Note-Taking-App.git
Cloning into 'Note-Taking-App'...
The authenticity of host 'github.com (20.205.243.166)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
git@github.com: Permission denied (publickey).
fatal: Could not read from remote repository.
```

- Installed the required project dependencies.

Go into the project folder:

```
cd repository
```

```
Anjan@anjan-vm:/var/www/Note-Taking-App$ ls
backend  frontend
```

Install dependencies (examples):

Node.js (Frontend/Backend)

- Install node

```
Anjan@anjan-vm:/var/www/Note-Taking-App$ sudo apt install nodejs
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcares2 libnode109 node-acorn node-busboy node-cjs-module-lexer node-undici node-xte
nd
  nodejs-doc
Suggested packages:
```

npm install

- Install npm inside VM

```
Anjan@anjan-vm:/var/www/Note-Taking-App/backend$ sudo chown -R Anjan:Anjan /var/www/Note
-Taking-App
Anjan@anjan-vm:/var/www/Note-Taking-App/backend$ ls -ld /var/www/Note-Taking-App
drwxr-xr-x 5 Anjan Anjan 4096 Sep 24 16:01 /var/www/Note-Taking-App
Anjan@anjan-vm:/var/www/Note-Taking-App/backend$ npm install
```

5. Configure NGINX

- Go to NGINX's site-available directory:

cd /etc/nginx/sites-available/

```
(venv) biraj@birajvm:/var/www/mystorage$ cd /etc/nginx/sites-available/
(venv) biraj@birajvm:/etc/nginx/sites-available$ ls
default
(venv) biraj@birajvm:/etc/nginx/sites-available$ |
```

- Create a config file for your project:

sudo nano project1.conf

Configured NGINX as a reverse proxy to forward requests to the application.

(Adjust port and folder paths as needed)

```
server {
    listen 80;
    server_name your_domain.com;
```

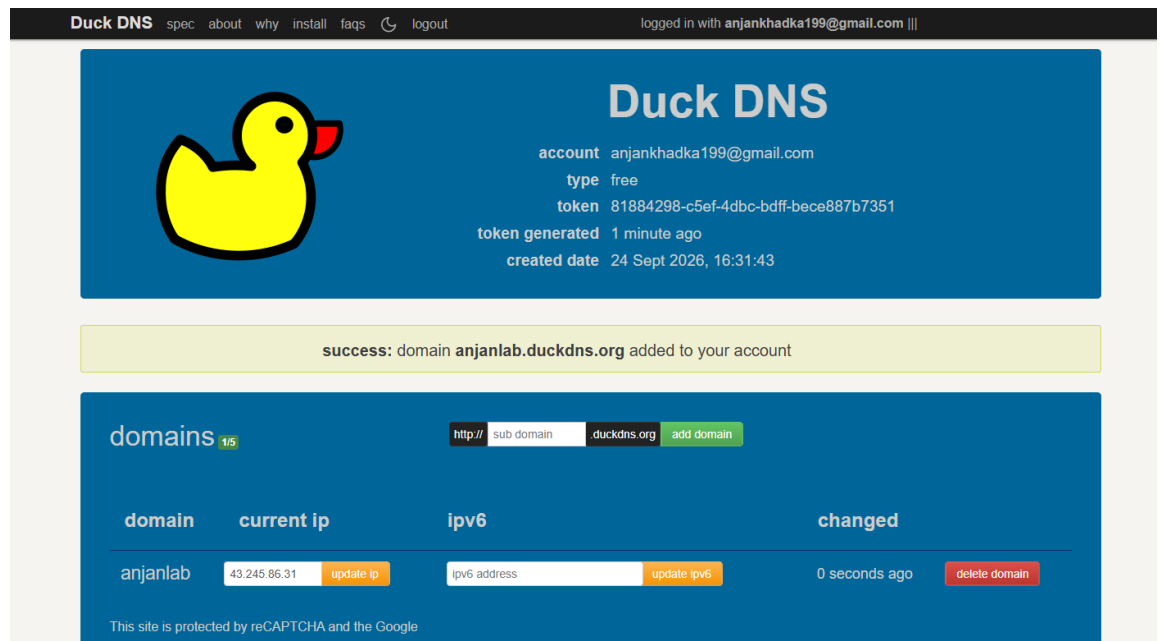
```

location / {
    proxy_pass http://127.0.0.1:8000; # Your app port
    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_set_header X-Forwarded-Proto $scheme;
}
}

```

Example

== We need a domain name for this purpose so we are using the free duckdns.org [domain](#) for the lab.



The screenshot shows the Duck DNS web interface. At the top, there's a navigation bar with links like 'spec', 'about', 'why', 'install', 'faqs', and a 'logout' button. The user is logged in with 'anjankhadka199@gmail.com'. The main header features a yellow duck logo and the 'Duck DNS' title. Below this, account details are listed: 'account: anjankhadka199@gmail.com', 'type: free', 'token: 81884298-c5ef-4dbc-bdff-bece887b7351', 'token generated: 1 minute ago', and 'created date: 24 Sept 2026, 16:31:43'. A success message states: 'success: domain anjanlab.duckdns.org added to your account'. The 'domains' section shows a table with one domain, 'anjanlab', with a current IP of '43.245.86.31' and an IPv6 address field. Buttons for 'update ip', 'update ipv6', and 'delete domain' are present. The footer mentions protection by reCAPTCHA and Google.

domain	current ip	ipv6	changed
anjanlab	43.245.86.31 update ip	ipv6 address update ipv6	0 seconds ago delete domain

```
Anjan@anjan-vm: /var/www/Note-Taking-App/frontend$ nano /etc/nginx/sites-available/project1.conf
GNU nano 7.2 project1.conf
server {
    listen 80;
    server_name anjanlab.duckdns.org;

    location /api/ {
        proxy_pass http://127.0.0.1:8000;
        proxy_http_version 1.1;

        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }

    location / {
        root /var/www/Note-Taking-App/frontend/dist;
        try_files $uri $uri/ /index.html;
    }
}
```

- Create a Symlink to Enable the Site

Link your config from sites-available to sites-enabled:

```
sudo ln -s /etc/nginx/sites-available/project1.conf /etc/nginx/sites-enabled/
```

```
Anjan@anjan-vm:/var/www/Note-Taking-App/frontend$ cd /etc/nginx/sites-available/
Anjan@anjan-vm:/etc/nginx/sites-available$ sudo nano project1.conf
Anjan@anjan-vm:/etc/nginx/sites-available$ sudo ln -s /etc/nginx/sites-available/project1.conf /etc/nginx/sites-enabled/
```

- Test NGINX Configuration

Check for syntax errors:

```
sudo nginx -t
```

```
Anjan@anjan-vm:/etc/nginx/sites-available$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
```

If everything is OK, proceed.

6. Reload or Restart NGINX

- Apply the changes:

sudo systemctl reload nginx

```
Anjan@anjan-vm:/etc/nginx/sites-available$ sudo systemctl reload nginx
Anjan@anjan-vm:/etc/nginx/sites-available$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-09-24 15:32:08 UTC; 1h 5min ago
     Docs: man:nginx(8)
   Process: 2170 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Process: 2172 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Process: 5766 ExecReload=/usr/sbin/nginx -g daemon on; master_process on; -s reload (code=exited, status=0/SUCCESS)
  Main PID: 2201 (nginx)
    Tasks: 3 (limit: 9432)
   Memory: 2.6M (peak: 5.2M)
      CPU: 33ms
   CGroup: /system.slice/nginx.service
           └─2201 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─5768 "nginx: worker process"
               └─5769 "nginx: worker process"

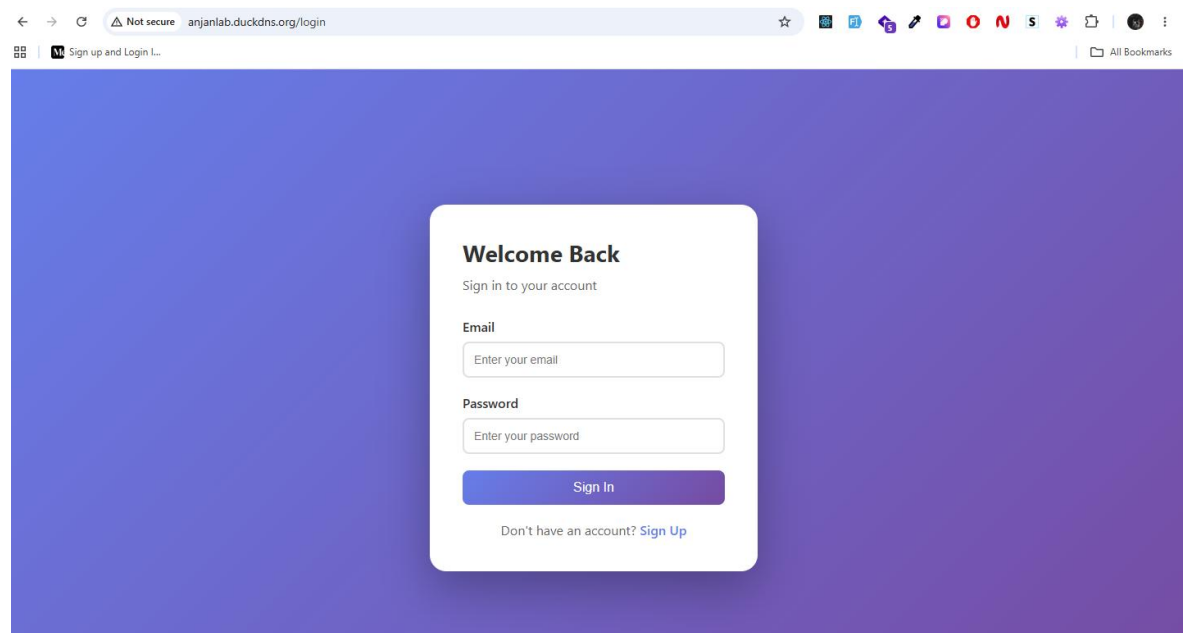
Sep 24 15:32:08 anjan-vm systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Sep 24 15:32:08 anjan-vm systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
Sep 24 16:37:04 anjan-vm systemd[1]: Reloading nginx.service - A high performance web server and a reverse proxy server...
Sep 24 16:37:04 anjan-vm nginx[5766]: 2026/09/24 16:37:04 [notice] 5766#5766: signal process started
Sep 24 16:37:04 anjan-vm systemd[1]: Reloaded nginx.service - A high performance web server and a reverse proxy server.
```

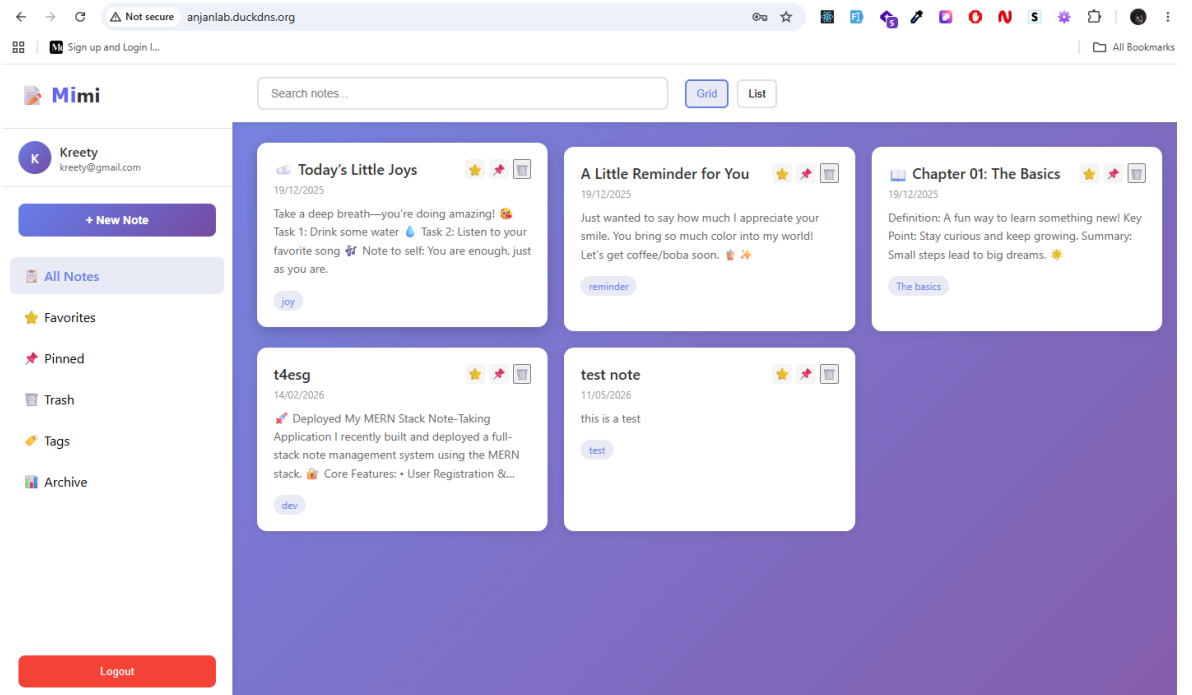
Or full restart:

sudo systemctl restart nginx

Check the domain you have configure to see apply changes on the domain you configured

<http://anjanlab.duckdns.org/>





Lab-5 Virtualization

Objective:

To install and configure virtualization software (VMware or VirtualBox) on a local machine and create multiple virtual machines.

Tools Used:

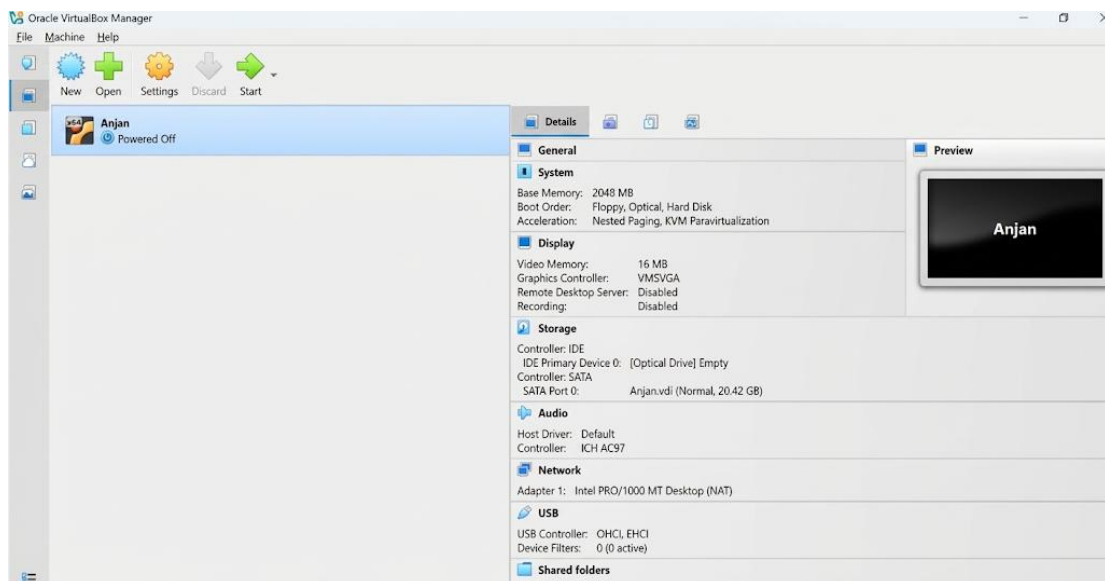
- VMware Workstation
- VirtualBox

In this practical, we will install and configure Linux Mint as a virtual machine (VM) using VirtualBox and VMware Workstation. Follow the step-by-step instructions provided for each tool.

1. Installing Linux Mint on VirtualBox

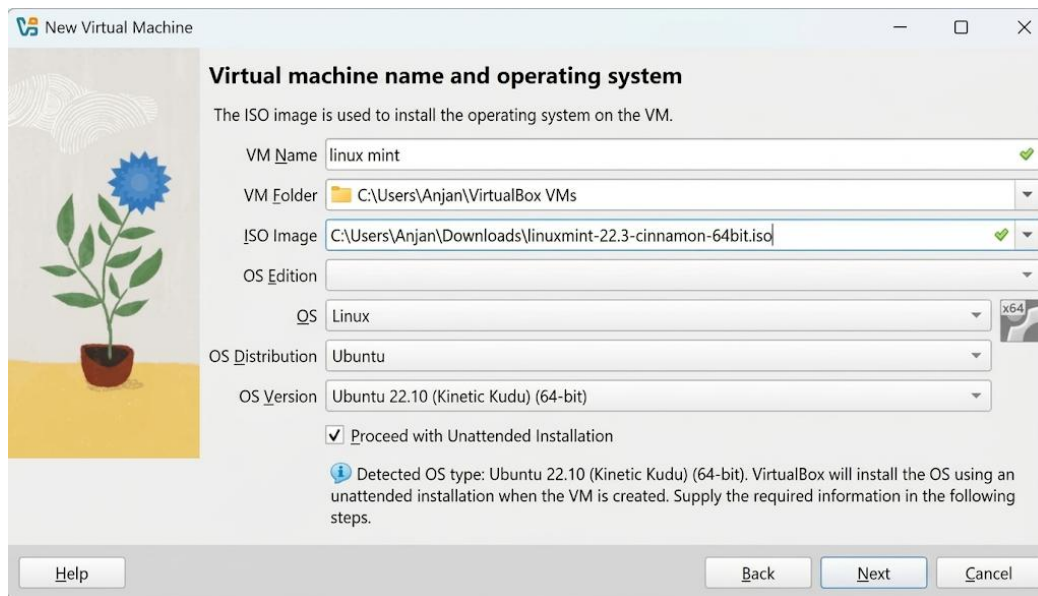
Step 1: Install VirtualBox

1. Download VirtualBox from the official website.
2. Install VirtualBox by following the on-screen instructions.
3. Open VirtualBox after installation.



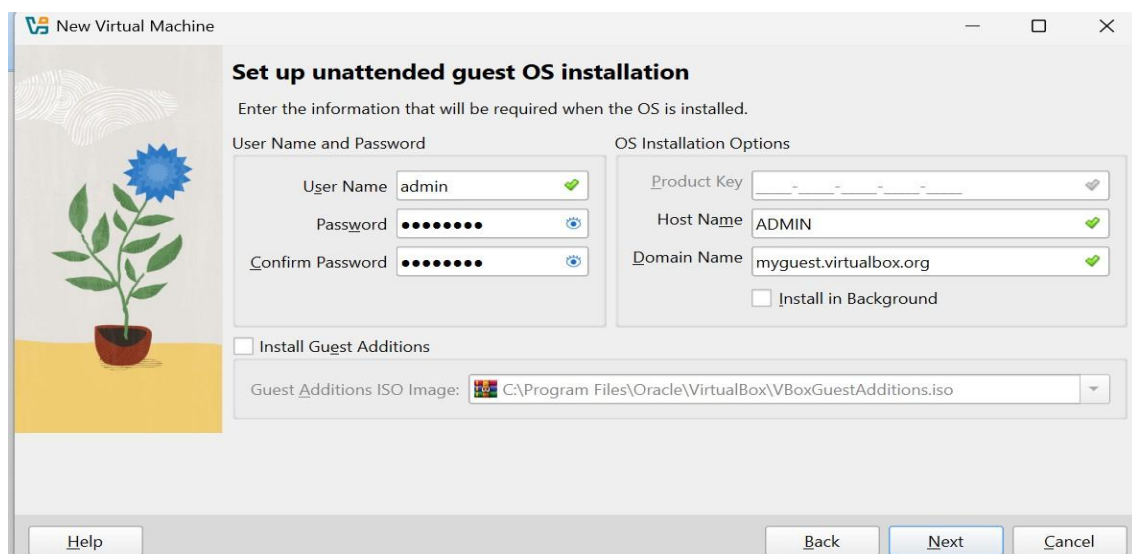
Step 2: Create a New Virtual Machine

- Click on "New" to create a new virtual machine.
- A pop-up window will appear. Configure the following:
 - a) Name: Enter a name for the VM (e.g., "Linux Mint").
 - b) Folder: Choose a location to store the VM files.
 - c) ISO Image: Select the Linux Mint ISO file.

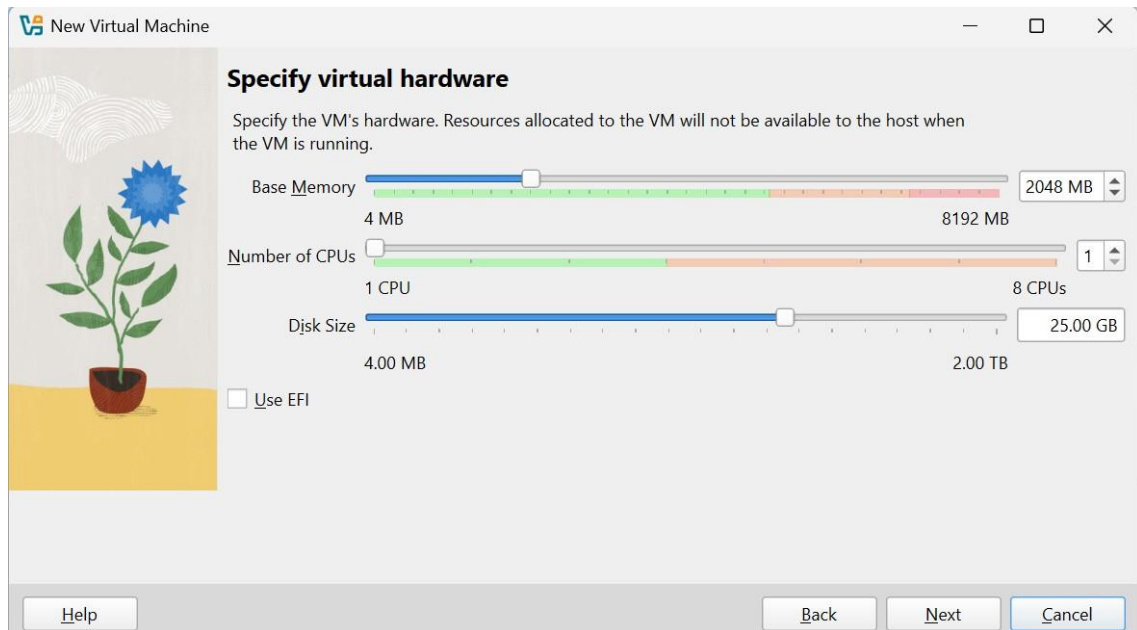


Step 3: Configure System Settings

- Set up username, password, and hostname for the VM.

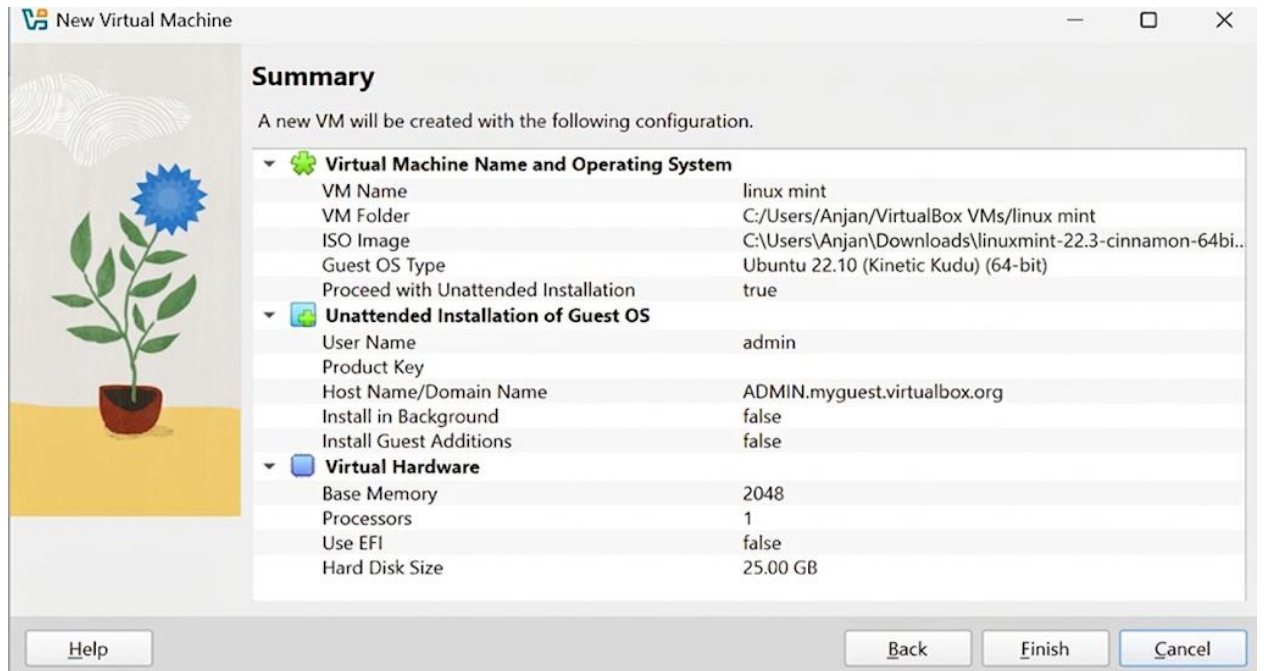


- Customize hardware resources based on system requirements:
 - a) Memory (RAM) allocation
 - b) Processor (CPU) cores
 - c) Storage capacity



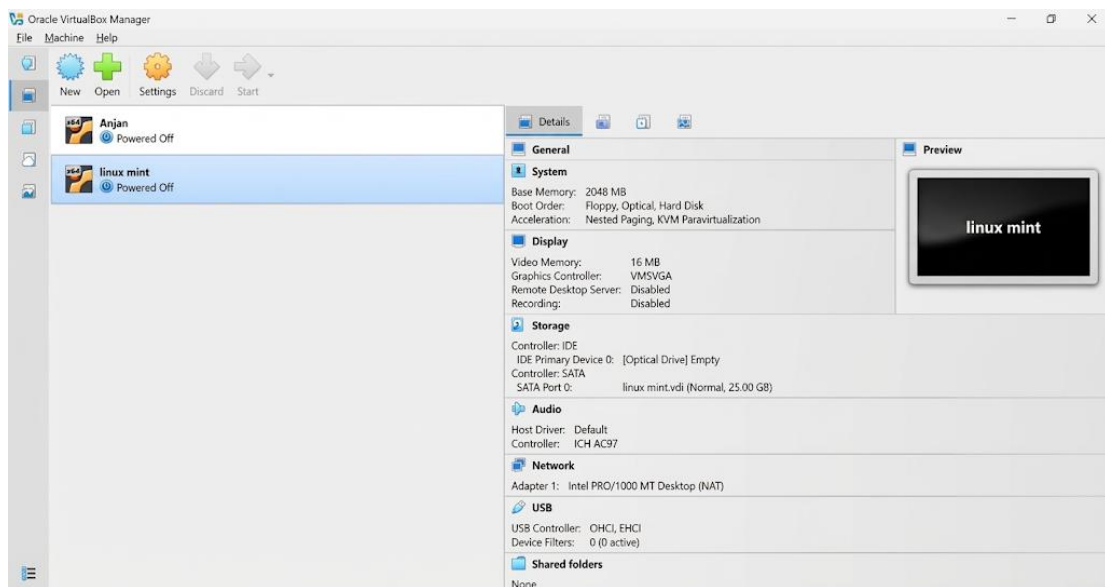
Step 4: Finalizing Setup

1. Review the summary of the virtual machine configuration.
2. Click "Finish" to complete the setup.

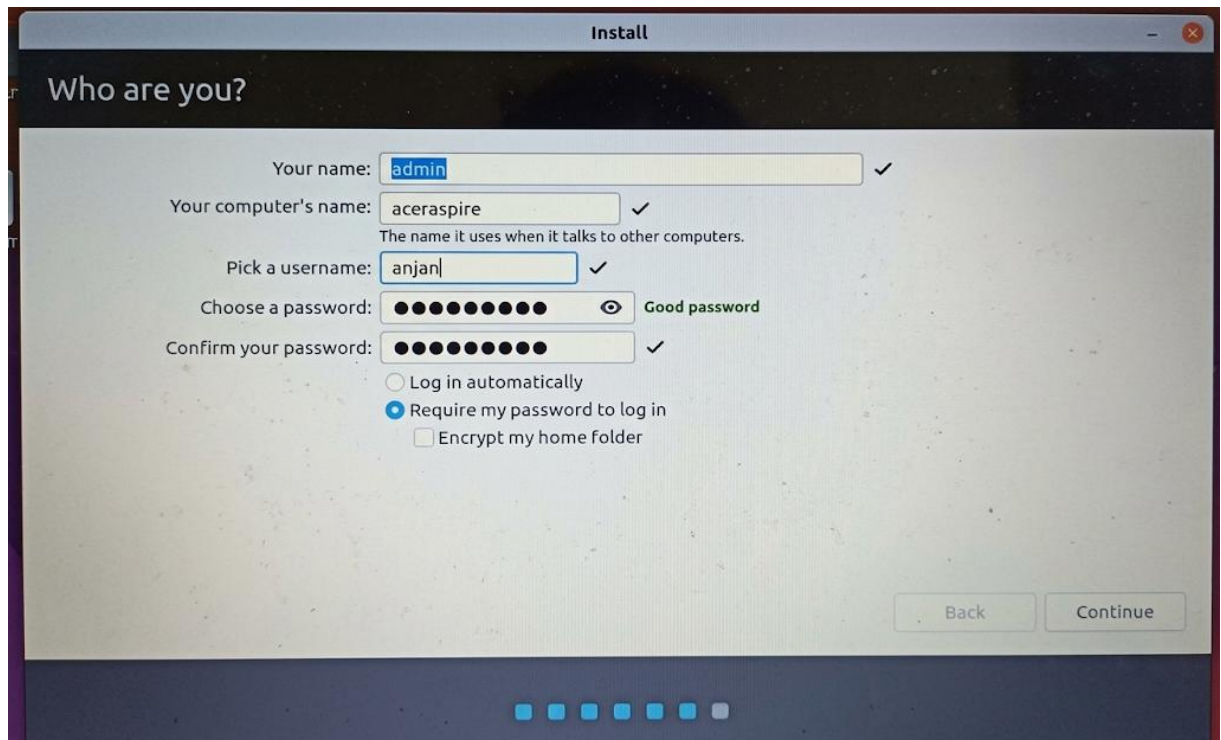


Step 5: Installing Linux Mint

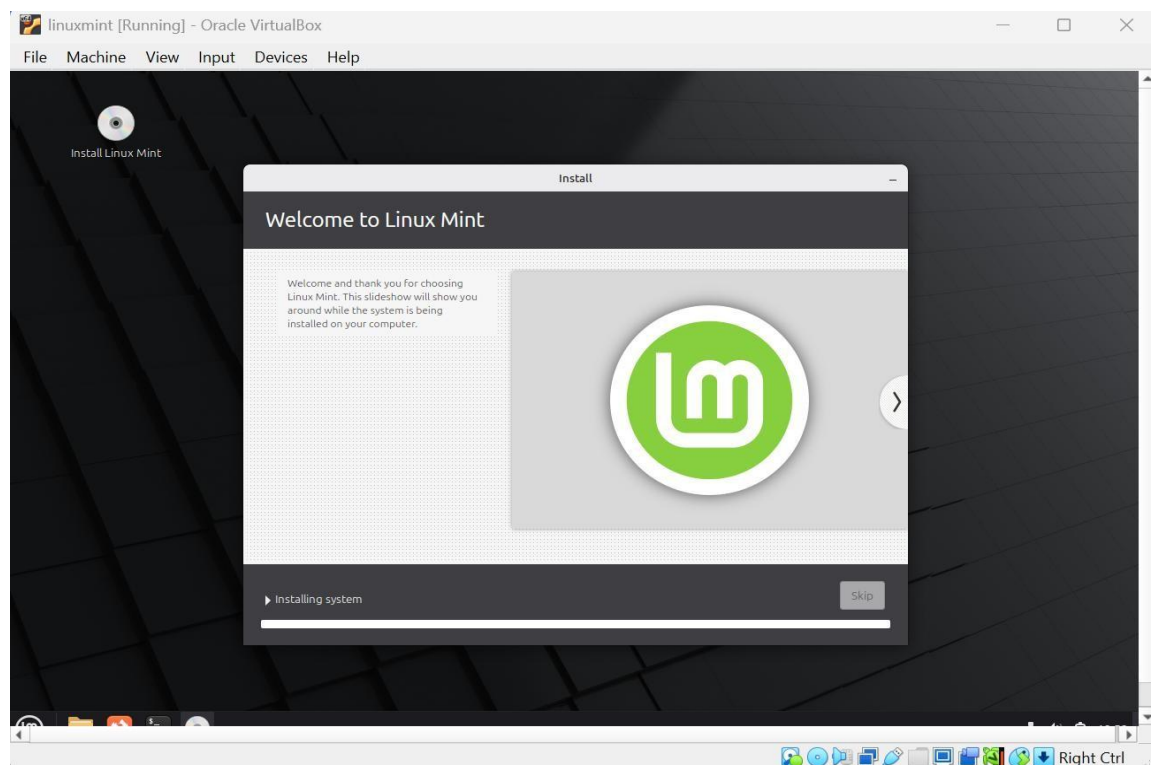
1. Start the virtual machine from the VirtualBox main interface.



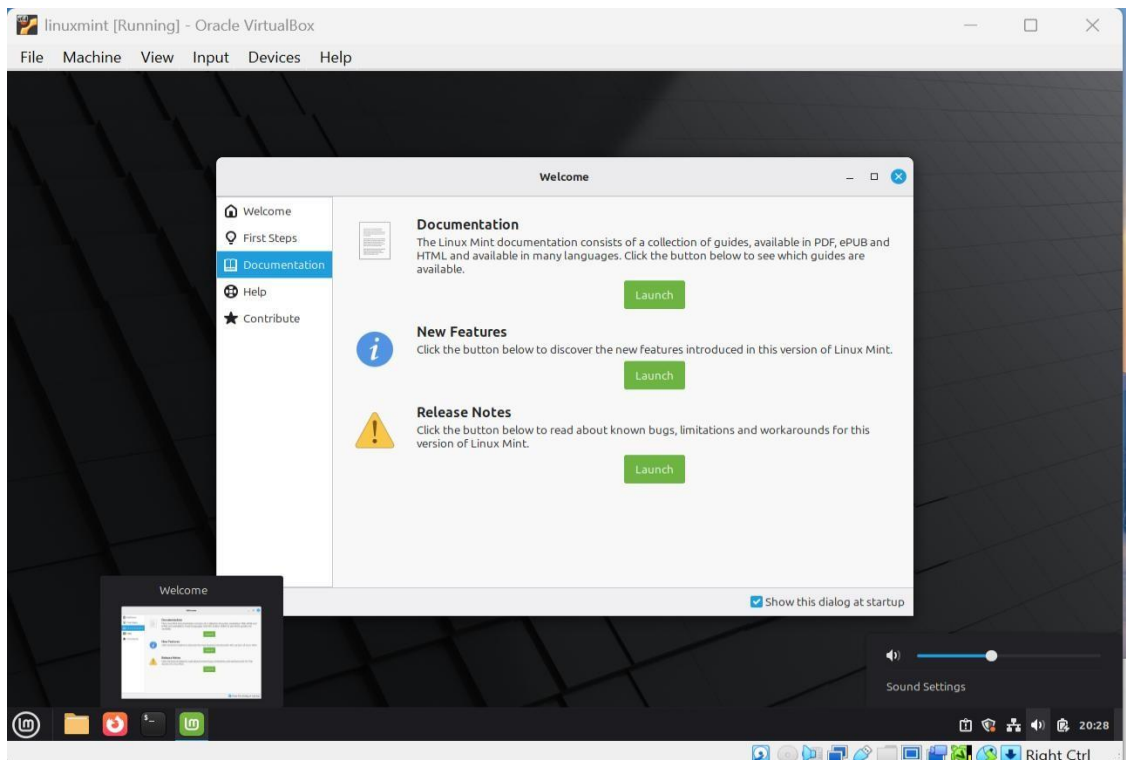
2. The installation process for Linux Mint will begin.
3. Follow on-screen instructions:
 - Pick a username and password.
 - Continue with the installation process.



4. Once completed, Linux Mint will successfully launch inside VirtualBox.



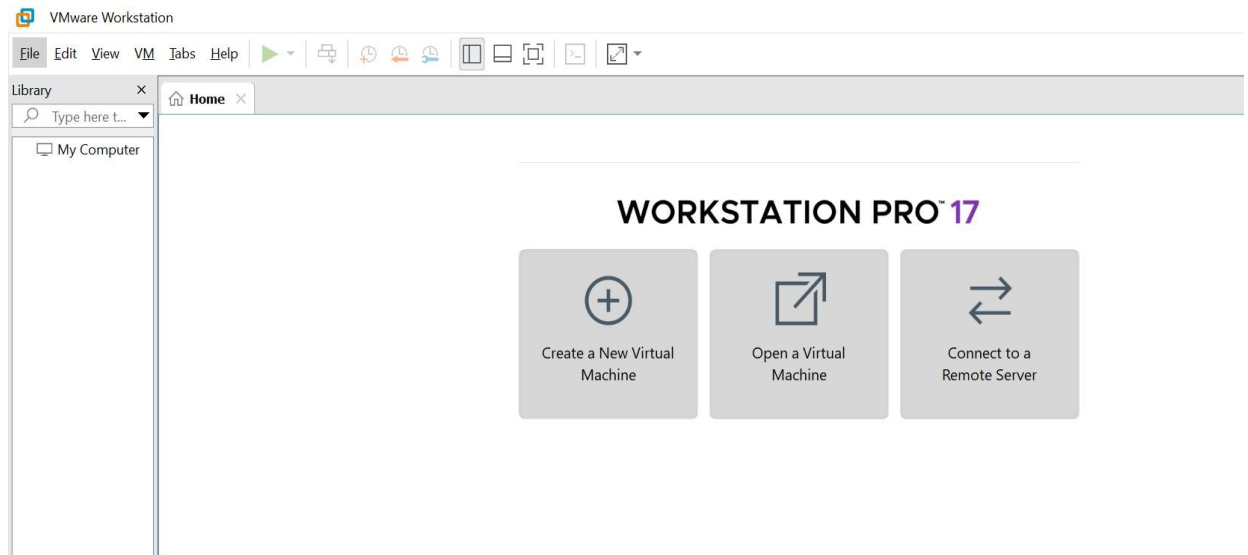
5. Successfully launch “Linux Mint” Operating system on Virtual Box



Lab-6 Installing Linux Mint on VMware Workstation

Step 1: Install VMware Workstation

- a) Download VMware Workstation from the official website.
- b) Install VMware Workstation and launch the application.



Step 2: Create a New Virtual Machine

1. Click on "Create a New Virtual Machine".
2. Choose the third option:
 - a. "I will install the operating system later" and click Next.

New Virtual Machine Wizard

Guest Operating System Installation
A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?

Install from:

☐ Installer disc:
No drives available

☐ Installer disc image file (iso):
C:\Users\Anjan\Downloads\linuxmint-22.3-cinnamon-64 Browse...

☒ I will install the operating system later.
The virtual machine will be created with a blank hard disk.

Help < Back Next > Cancel

Step 3: Configure VM Settings

1. Select the Guest Operating System:
 - Choose Linux as the OS type.
 - If Linux Mint Cinnamon is not listed, select Ubuntu 64-bit (closest version).
2. Click Next.

New Virtual Machine Wizard

Select a Guest Operating System
Which operating system will be installed on this virtual machine?

Guest operating system

☐ Microsoft Windows

☒ Linux

☐ VMware ESX

☐ Other

Version

Ubuntu 64-bit

Help < Back Next > Cancel

Step 4: Set VM Name and Storage Location

1. Enter a name for the virtual machine (e.g., "linuxmint").
2. Specify a location to save the VM files.
3. Click Next.

New Virtual Machine Wizard

Name the Virtual Machine
What name would you like to use for this virtual machine?

Virtual machine name:

Location:

The default location can be changed at Edit > Preferences.

< Back Next > Cancel

Step 5: Specify Hard Disk Capacity

1. Set the desired disk size (recommended: 20GB or more).
2. Click Next.

The screenshot shows a window titled "New Virtual Machine Wizard" with a close button (X) in the top right corner. The main heading is "Specify Disk Capacity" with the subtext "How large do you want this disk to be?". Below this, a paragraph explains: "The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine." A text input field for "Maximum disk size (GB):" contains the value "20.0" and has up/down arrow buttons. Below the input field, it says "Recommended size for Ubuntu 64-bit: 20 GB". There are two radio button options: "Store virtual disk as a single file" (unselected) and "Split virtual disk into multiple files" (selected). A descriptive text for the selected option reads: "Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks." At the bottom, there are four buttons: "Help", "< Back", "Next >" (highlighted with a blue border), and "Cancel".

New Virtual Machine Wizard

Specify Disk Capacity
How large do you want this disk to be?

The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

Maximum disk size (GB):

Recommended size for Ubuntu 64-bit: 20 GB

☐ Store virtual disk as a single file

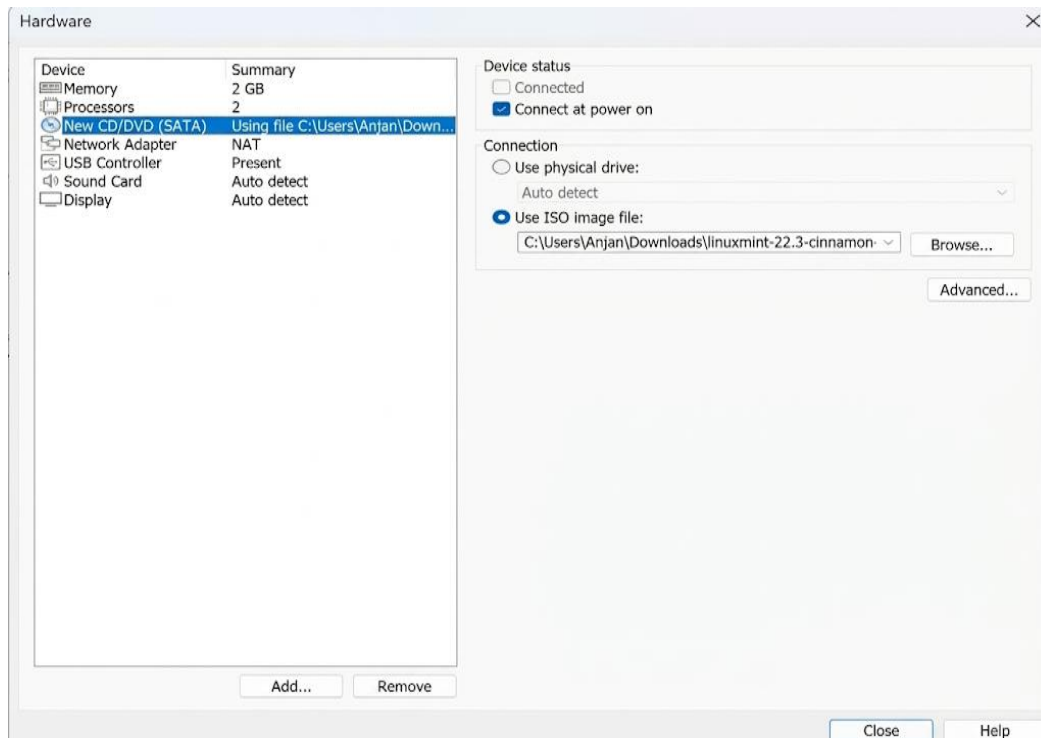
☒ Split virtual disk into multiple files

Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

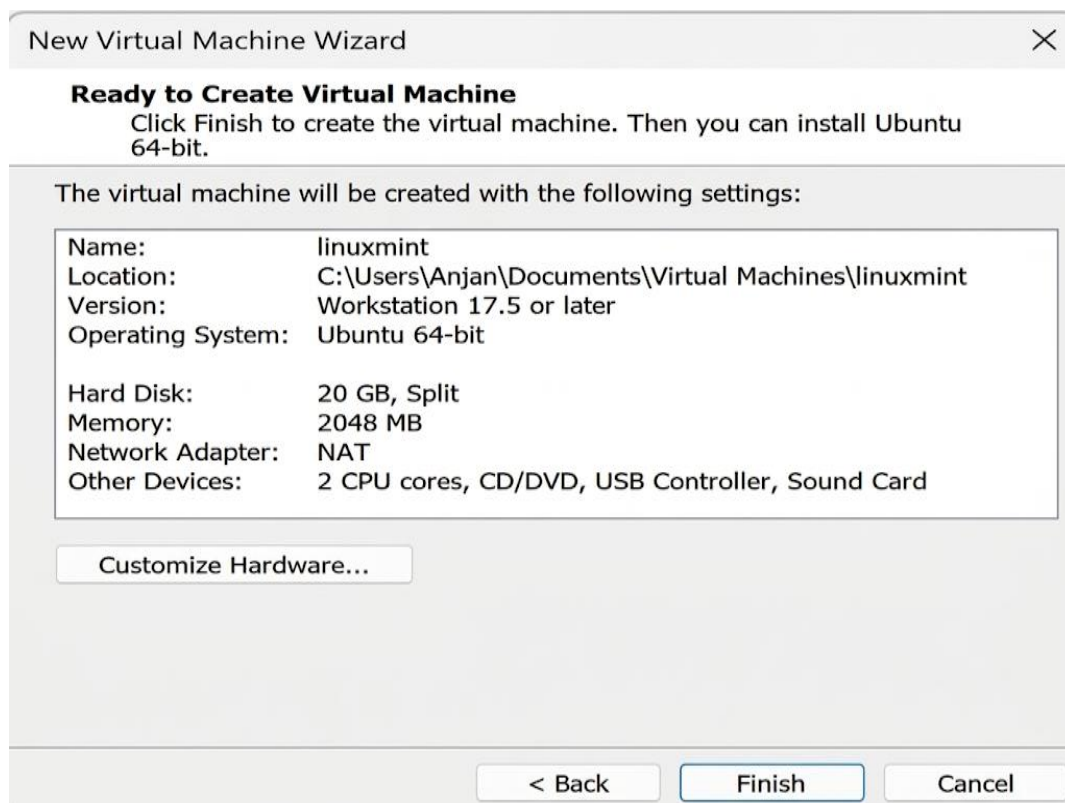
Help < Back Next > Cancel

Step 6: Customize Hardware Configuration

1. Adjust Memory (RAM), Processors, and other hardware settings.
2. Locate the ISO image file:
 - Go to New CD/DVD (SATA).
 - Browse and select the Linux Mint ISO file.
 - Click Close.
3. Click Finish to complete the setup.

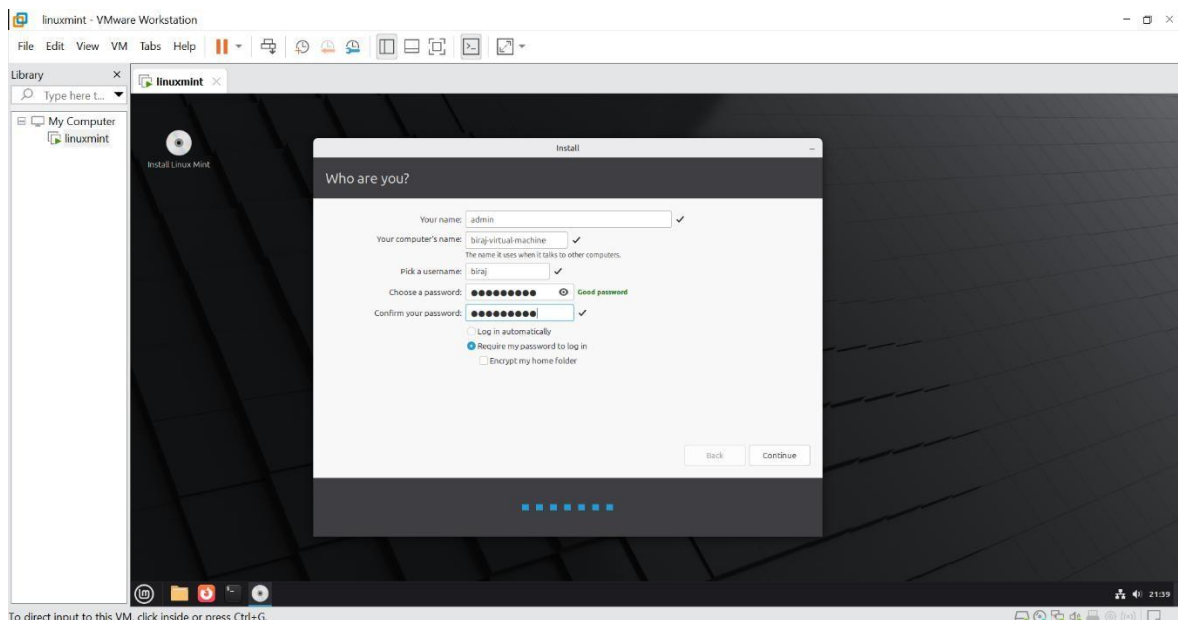
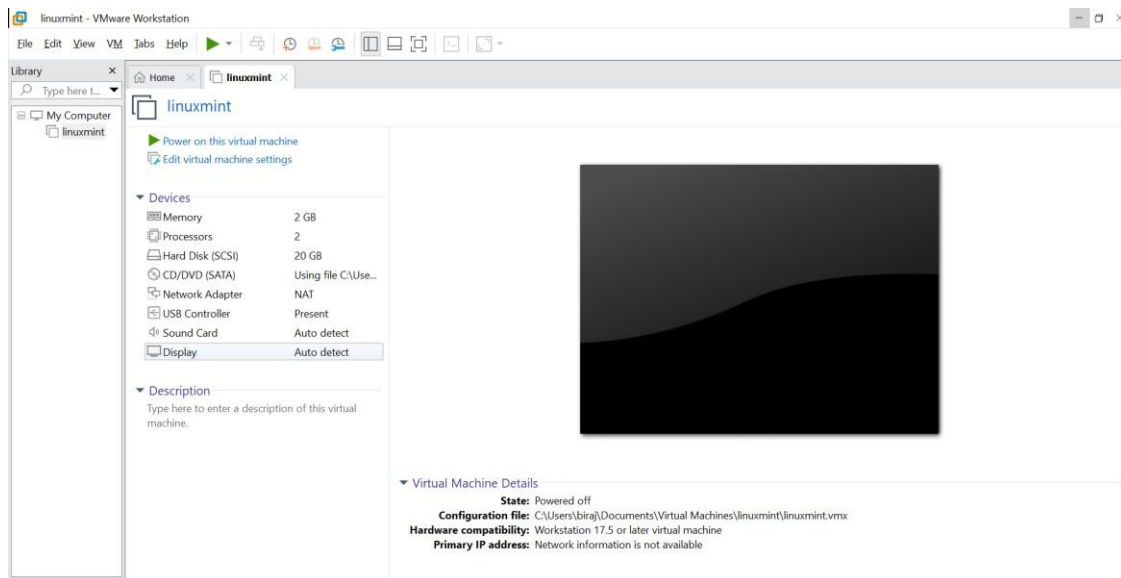


4. Do finish.



Step 7: Start the Virtual Machine

1. Select the newly created VM from the list.
2. Click "Play Virtual Machine" to start it.
3. The Linux Mint installation process will begin.
4. Complete the setup by creating a username and password



Once the installation is finished, Linux Mint will successfully launch on VMware Workstation.

